

Optimal Order Selection via the Dinkelbach Method

Gustavo Machado Leal, Rafael Castro, Humberto Longo

Instituto de Informática, Universidade Federal de Goiás

Alameda Palmeiras, Quadra D, Campus Samambaia, CEP 74690-900, Goiânia, Goiás, Brazil

{gustavoleal, rafael_castro}@discente.ufg.br, longo@ufg.br

ABSTRACT

This article study the Optimal Order Selection problem posed in the SBPO 2025 challenge, in which one must choose a subset of customer orders and warehouse aisles so as to maximize the ratio of total items picked to the number of aisles activated, under given lower and upper bounds on wave size. A pure Dinkelbach-based approach that reformulates the fractional objective into a sequence of parametric mixed-integer programs is proposed, each solved to optimality by CPLEX. On 35 official benchmark instances (20 from Dataset A, 15 from Dataset B), the method achieves optimal ratios on 25 cases, and respects the 10 min time limit.

KEYWORDS. Fractional Programming, Mixed Integer Programming, Dinkelbach's Method, CPLEX, Optimal Order Selection

1. Introduction

The Optimal Order Selection problem, introduced in the SBPO 2025 challenge by Mercado Libre [Mercado Libre SBPO Committee, 2025], requires forming “picking wave” that maximize the average number of items picked per aisle activated, while respecting lower and upper bounds on total demand and ensuring full coverage of every item in the selected orders. This problem, which generalizes the NP-hard Set Cover problem [Karp, 1972], poses significant computational challenges at industrial scale. In this work a Dinkelbach-based algorithm is developed and analysed, which that directly tackles the fractional objective via a sequence of parametric MIPs, leveraging CPLEX's performance and warm-start capabilities to deliver provably optimal or near-optimal solutions within tight time budgets.

First, in Section 2 the Optimal Order Selection problem is formalized as defined in the SBPO 2025 challenge. Section 3 then establishes its NP-hardness. Next, Section 4 introduces the original fractional programming model. In Section 5 Dinkelbach's Method is presented, and Section 6 shows how the problem is reformulated into a sequence of parametric MIPs and gives pseudocode for the algorithm. Section 7 reports computational experiments on the official Datasets A and B. Finally, Section 8 summarizes the results and outlines future research directions.

2. Problem Description

Given a backlog O of customer orders and a set A of warehouse aisles, select a subset $O' \subseteq O$ and aisles $A' \subseteq A$ to maximize average units per aisle. Let:

- I be the set of all items across orders;
- I_o the items in order o ;
- A_i the aisles containing item i ;
- u_{oi} quantity of item i in order o ;
- u_{ai} capacity of item i in aisle a ;

- LB, UB lower/upper wave size bounds.

Feasible waves satisfy:

$$LB \leq \sum_{o \in O'} \sum_{i \in I_o} u_{oi} \leq UB,$$

$$\sum_{o \in O'} u_{oi} \leq \sum_{a \in A'} u_{ai}, \quad \forall i \in I.$$

3. Reduction from Set Cover

In this section is shown that the *Optimal Order Selection Problem* generalizes the classical *Set Cover Problem (SCP)*. The decision variant of the SCP problem is one of the 21 classical problems presented by Karp [1972]. The SCP is defined as: given a universe $U = \{e_1, e_2, \dots, e_n\}$ and a collection of subsets $\mathcal{S} = \{S_1, S_2, \dots, S_m\}$ such that $S_j \subseteq U$, determine the smallest number of subsets whose union equals U . We reduce an instance of SCP to a restricted version of the order selection problem by applying these constraints:

1. Fix the wave size bounds to force all orders to be included:

$$LB = UB = \sum_{o \in O} \sum_{i \in I_o} u_{oi} \Rightarrow O' = O$$

2. Each item appears in exactly one order and is requested in quantity 1;
3. The objective reduces to minimizing the number of aisles used:

$$\max \frac{\sum_{o \in O'} \sum_{i \in I_o} u_{oi}}{|A'|} \Rightarrow \min |A'| \text{ (since numerator is fixed)}$$

Under this construction:

- The universe U becomes the set of items I .
- Each aisle $a \in A$ becomes a subset of items $A_a \subseteq I$ it contains.
- Finding the minimal set $A' \subseteq A$ such that all items in I are covered corresponds exactly to a solution to the Set Cover Problem.

Therefore, the *Optimal Order Selection Problem* generalizes SCP and is thus NP-hard. As a result, it becomes essential to design and evaluate efficient heuristic algorithms capable of quickly producing high-quality solutions. These methods offer a practical balance between solution quality and runtime performance, especially when dealing with large-scale industrial data and operational constraints.

4. Model Formulation

Let binary variables x_o indicate whether order $o \in O$ is selected, and y_a indicate whether aisle $a \in A$ is selected. The objective is to maximize the number of items collected per aisle used:

$$\max \frac{\sum_{o \in O} \sum_{i \in I_o} u_{oi} \cdot x_o}{\sum_{a \in A} y_a}$$

Subject to:

$$\begin{aligned}
 LB &\leq \sum_{o \in O} \sum_{i \in I_o} u_{oi} \cdot x_o \leq UB \\
 \sum_{o \in O} u_{oi} \cdot x_o &\leq \sum_{a \in A} u_{ai} \cdot y_a, \quad \forall i \in I \\
 x_o &\in \{0, 1\}, \quad \forall o \in O \\
 y_a &\in \{0, 1\}, \quad \forall a \in A
 \end{aligned}$$

The objective function is fractional and non-linear. When represented directly, it becomes a mixed-integer quadratic fractional program (MIQFP), which is significantly more difficult to solve than standard linear MIPs. Moreover, many classical heuristics are not designed to handle non-linear objectives, particularly those with variable denominators.

5. Dinkelbach's Method

In this case, the main problem of the model is the objective function, thus, applying some type of modification in it, leading to a MIP model would drastically increase the efficiency of any method. As originally presented by Dinkelbach [1967], Dinkelbach's Method solves a sequence of parametric subproblems with linear convergence and quadratic local behavior.

To handle the ratio $f(x)/g(y)$, Dinkelbach's Method iteratively solves:

$$\max_{x,y} f(x) - \alpha_k g(y),$$

updating

$$\alpha_{k+1} = \frac{f(x_k^*)}{g(y_k^*)},$$

until

$$|f(x_k^*) - \alpha_k g(y_k^*)| < \epsilon.$$

As proven by Dinkelbach [1967], this procedure generates a sequence $\{\alpha_k\}$ that is:

- **Monotonic:** $\alpha_{k+1} \geq \alpha_k$ whenever $g(x_k^*) > 0$; and
- **Convergent** to the global optimum α^* of the original problem.

No additional line search or penalty parameters are required, making the method straightforward to implement. An equivalent characterization is that $f(x) - \alpha g(y)$ has optimal objective value zero if and only if $\alpha = \alpha^*$. Thus, checking for $f(x_k^*) - \alpha_k g(x_k^*) = 0$ serves both as a convergence and optimality criterion. In practice, the algorithm converged under the stopping tolerance $|f(x_k^*) - \alpha_k g(x_k^*)| < 10^{-3}$. or under the time restriction (10 minutes).

6. Reformulated Model via Dinkelbach

The new model is simple, updating the objective function to the format of Dinkelbach's Method. At iteration k , solve the MIP instance:

$$\max_{x,y} \sum_{o,i} u_{oi} x_o - \alpha_k \sum_a y_a, \quad (1)$$

$$\text{s.t. } LB \leq \sum_{o,i} u_{oi} x_o \leq UB, \quad (2)$$

$$\sum_o u_{oi} x_o \leq \sum_a u_{ai} y_a, \quad \forall i, \quad (3)$$

$$x_o, y_a \in \{0, 1\}. \quad (4)$$

Update α_{k+1} and repeat.

Importantly, the decision variable structure and constraints remain unchanged. This allows efficient reuse of warm-start solutions across iterations.

Pseudocode: Iterative Dinkelbach Optimization

The following pseudocode presents the algorithm used in the experiment:

Input:

Initial $\alpha_{k+1} \leftarrow 0$

Number of iterations $T = 15$

For $k = 1$ to T do:

1. Construct the relaxed objective:

$$\sum_o u_{oi} \cdot x_o - \alpha_k \cdot \sum_a y_a$$

2. Call CPLEX solver to solve the relaxed model:

- Use previous solution - Maintain all constraints (bounds, binary conditions)

3. Compute g :

$$g \leftarrow \frac{\sum_o u_{oi} \cdot x_o}{\sum_a u_{ai} \cdot y_a}$$

4. Update alpha:

$$\alpha_{k+1} \leftarrow g$$

7. Computational Results

The relaxed model was implemented and solved using IBM ILOG CPLEX 22.11. To maximize efficiency, only the objective coefficients were modified between iterations. Binary variables x_o and y_a , as well as the full constraint structure (e.g., wave size bounds and integrality), were preserved unchanged.

The solutions obtained in one iteration were reused as initial feasible solutions in the following one. This strategy significantly reduced solving times and maintained solution quality. In our experiments, CPLEX maintained feasible primal solutions across all iterations, even as the objective function evolved.

Experiments were conducted under the 10-minute per-instance time limit as specified by the SBPO 2025 challenge, on a Windows 11 workstation equipped with a 13th Gen Intel Core i9-13900H processor (2.60 GHz) and 16 GB of RAM. Table 1 reports performance on 35 instances (20 from Dataset A, 15 from Dataset B). The columns are:

- **Instance:** Unique instance code (e.g., A01, B05).
- **#Orders:** Number of orders in the instance.
- **#Aisles:** Number of aisles in the instance.
- **Ratio:** Final value of f/g obtained by Dinkelbach's Method.
- **Optimal:** "Yes" if the computed ratio matches the known optimal; "No" otherwise.
- **Time (s):** CPU time in seconds until convergence (within the 10-minute limit).

Instance	#Orders	#Aisles	Ratio	Optimal	Time (s)
A01	61	116	15.0	Yes	0s
A02	7	33	2.0	Yes	0s
A03	82	124	12.0	Yes	0s
A04	16	91	3.5	Yes	0s
A05	2625	161	177.88	Yes	131s
A06	10341	184	691.0	Yes	22s
A07	8320	180	392.25	Yes	13s
A08	2185	168	162.94	Yes	67s
A09	70	304	4.4	Yes	0s
A10	1602	383	16.35	No	158s
A11	1029	375	16.85	Yes	1.2
A12	133	342	11.25	Yes	0s
A13	8375	413	117.4	Yes	39s
A14	12402	413	181.64	Yes	367s
A15	7367	402	149.3	Yes	25s
A16	1108	88	85.0	Yes	0s
A17	417	83	36.5	Yes	0s
A18	2682	90	117.2	Yes	6s
A19	2257	134	202.0	Yes	2s
A20	5	5	5.0	Yes	0s
B01	11321	185	631.5	Yes	10s
B02	2162	165	180.6	Yes	51s
B03	2070	164	149.6	Yes	154s
B04	2448	163	147.47	No	216s
B05	6468	179	368.0	Yes	9s
B06	1857	165	83.42	No	538s
B07	1841	165	108.0	Yes	1s
B08	12334	398	227.1	No	589s
B09	5581	417	82.49	No	62s
B10	14952	482	444.0	Yes	158s
B11	45112	482	258.12	No	589s
B12	28263	483	561.27	No	589s
B13	38214	483	702.0	No	589s
B14	38202	483	679.1	No	589s
B15	11541	398	162.0	No	328s

Table 1: Computational results on Datasets A and B.

8. Conclusion

The results presented in this article demonstrate that a pure Dinkelbach-Method approach efficiently and reliably solves the Optimal Order Selection problem. On 35 challenging instances, the proposed algorithm achieved the optimal results in 25, all under the 10 minutes restriction. These findings confirm that Dinkelbach's Method, when combined with state-of-the-art MIP solvers and warm-start strategies, provides a powerful tool for large-scale fractional optimization in warehousing and distribution. Future work will explore hybrid schemes, integrating Lagrangian relaxation, advanced subgradient techniques, and decomposition methods, to further accelerate convergence and enhance primal feasibility in the most demanding industrial scenarios.

8.1. Future Works

While this paper focuses exclusively on the Dinkelbach Method for fractional programming, a natural extension is to integrate *Lagrangian Relaxation* to handle particularly challenging coverage constraints. Future research directions include:

- Developing a hybrid Dinkelbach and Lagrange algorithm that dynamically relaxes item coverage constraints via dual multipliers to improve primal feasibility.
- Exploring *partial relaxation* strategies, selectively applying Lagrange multipliers only to the most violated constraints, to speed convergence.
- Incorporating advanced subgradient or bundle methods to accelerate dual convergence within each Dinkelbach iteration.

References

- Dinkelbach, W. (1967). On nonlinear fractional programming. *Management Science*, 13(7):492–498.
- Karp, R. M. (1972). Reducibility among combinatorial problems. In Miller, R. E. and Thatcher, J. W., editors, *Complexity of Computer Computations*, p. 85–103. Plenum Press.
- Mercado Libre SBPO Committee (2025). Sbp0 2025 challenge description and rules. Technical report, Mercado Libre. Available at the SBPO 2025 official website.